Applied Energy 235 (2019) 332–345

Contents lists available at ScienceDirect

Applied Energy

journal homepage: www.elsevier.com/locate/apenergy

An interactive planning model for sustainable urban water and energy supply

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HIGHLIGHTS

- Surplus photovoltaics output and grid electricity provide energy for water supply.
- Sustainable planning using a two-level mixed integer linear programming model.
- Short-term operational constraints are integrated with yearly planning model.
- Techno-economic and subjectivity criteria affect photovoltaic uptake capacity.
- Scenarios and sensitivity analysis are done for the case-study of Perth, Australia.

ARTICLE INFO

Keywords:
Photovoltaic system
Grid electricity
Urban water supply
Short-term scheduling
Long-term planning

ABSTRACT

An interactive multi-period planning model is presented for sustainable urban water and energy supply, taking into account surplus output from grid-connected residential photovoltaics as a part of the water-related energy mix. The two-level mixed integer linear model finds the optimal strategic and operational decisions for a desalination-based water supply system driven by hybrid energy sources and determines the evolution of the potential capacity of a renewable energy technology over the planning horizon. It considers demands, supply systems configuration, resources capacities and electricity tariffs as well as economic, subjectivity and technical criteria for uptaking rooftop photovoltaic systems. The model was then applied to Perth (Australia) and solved for alternative scenarios. The results show operational flexibility and decentralised planning of the integrated system lead to $251,515,132 less discounted total cost over centralised water supply system operated in fixed mode. They also indicate that decentralised scenario results in 42,765.1 kW higher potential photovoltaics uptake capacity on average in each year over the planning horizon in the case study area compared to centralised scenarios. However, based on the results of the sensitivity analysis, the selection of this scenario as the best alternative highly depends on the parameters values associated with subjectivity criterion and operational and maintenance cost of flexible mode of operation.

1. Introduction

Diminishing water and energy resources along with more than ever increasing demand question the security of the future supply especially in cities, the centres of population and economic developments. The situation becomes more critical due to the uncertainties facing policy and decision makers at all levels of urban water and energy supply as well as the lack of coherence between the water and energy sectors.

Water is applied in fuel extraction, transportation, production, refining and processing as well as power generation and power plant cooling. On the other hand, Energy is used in water supply systems from water extraction and pumping to purification and distribution. The latter has an important impact on energy sector when sustainability of urban water supply cannot be ensured but through employing some of energy intensive technologies such as desalination. Construction, operation and expansion of these alternatives impose a significant pressure on energy resources. In fact, in some cases it has been reported that desalination and long-haul transfer are up to twenty three times more energy intensive compared to conventional surface water treatment [1].

Such interactions highlight the necessity of long-term supply planning for each of these commodities in the context of the integrated water and energy management and calls for optimisation modelling

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https://doi.org/10.1016/j.apenergy.2018.10.128
Received 16 August 2018; Received in revised form 26 October 2018; Accepted 30 October 2018
Available online 06 November 2018
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frameworks that consider these interlinks for efficient use of existing resources. In this regard, there are several questions that optimisation models need to answer: (1) How are strategic decisions affected by real-time dispatch of water and energy systems? (2) What are the effects of optimal decisions in each sector on the other temporally and spatially? (3) What are the impacts of social and economic uncertainties on the optimal decisions?

To address these challenges, there is a need to capture the water and energy linkage considering the modelling frameworks commonly used in each sector individually and then to compromise and adjust these conventional modelling approaches. In energy sector, there are numerous studies which have focused on optimisation of power generation expansion planning (GEP) problems. Sadeghi et al. [2] provided a detailed overview on GEP problems from different aspects of technological advances, climate change, control strategies and polices as well as applied optimisation models. Guerra et al. [3] developed a mixed integer linear programming (MILP) model for generation and transmission capacity expansion planning of an interconnected power system. Oree et al. [4] reviewed the optimisation techniques in GEP problems with focus on systems integrated with renewable energy sources. With respect to high penetration of renewable energies in GEP problems, Flores-Quiroz et al. [5] and Luz et al. [6] proposed a MILP and multi-objective optimisation model, respectively. In a recent study, Min et al. [7] used a stochastic optimisation approach for long-term capacity expansion of a power system integrated with large-size renewable energy technologies. Other studies have addressed long-term energy mix planning using different optimisation approaches such as stochastic programming [8], MILP model [9] and multi-objective techniques [10].

In water sector, optimisation modelling has been also used for long-term planning of the supply system. Tayfur [11] presented a comprehensive review on the methods used in water resources planning, engineering and management. Adeyemo and Stretch [12] reviewed the application of hybrid evolutionary algorithms in optimising reservoirs. Balekelayi and Tesfamariam [13] provided an overview on techniques employed in water distribution system including rehabilitation. Recently, Marques et al. [14] applied an enhanced simulated annealing algorithm to address a many-objective optimisation design problem in water distribution system in a long-term planning horizon. In another study, Ghelichi et al. [15] proposed a deterministic multi-echelon multi-period model for long-term planning of a municipal water distribution network. They used a two-stage scenario-based stochastic programming to cope with parameters uncertainties such as demand. In regard to desalination-based water supply system, Al-Nory et al. [16] introduced the concept of “desalination supply chain” and presented a methodology to address various aspects which need to be considered in its long-term planning. Using MILP model, Saif and Almansoori [17] and Shahabi et al. [18] presented multi-period construction and capacity expansion planning of a desalination-based water supply system at regional and city scales, respectively.

There are also several studies in the literature focusing on optimal long-term planning of integrated water and energy supply systems simultaneously. Segurado et al. [19] obtained optimal investment and operational decisions for an integrated system consisting of wind-powered desalination plant, reservoir, pumped hydro storage and fossil fuel based generators using H2RES simulation tool and iteration technique. In their next study [20], they focused more on operational strategy and size optimisation based on the parameters values which were foreseen for 2020. They applied simulation along with a derivative free multi-objective optimisation method. From the Pareto optimal set, several solutions could be chosen based on the selected criteria. Using INFINIT (interdependent network flows with induced internal transformation) mode, Ishimatsu et al. [21] conducted a single facility analysis to find the best location of a desalination plant driven by the renewable energy as its primary source. Considering the geographical aspect of the problem, they used a MILP- based model to minimise both total cost and CO_2 emissions to find a solution at national scale (Saudi Arabia) in 2030. Novosel et al. [22] employed the EnergyPlan model to investigate the effect of different but fixed configurations of an integrated desalinated water and energy supply system on the share of renewable energy as well as their benefits on the Jordanian energy supply system in 2020, 2030 and 2050. In another study, Caldera et al. [23] used the linear programming (LP)-based model integrated with levelised cost (LC) analysis to indicate the benefit of providing the total desalinated water demand solely with renewable energy in 2030. These models address one-period optimisation problems to indicate the economic and/or sustainability benefits of the optimal system in one/several target year(s) in the future. They act as if decisions made in the future are decoupled from those in the previous periods. Therefore, they cannot be accurate representatives of real world problems where planning over time is conducted by partitioning time into a number of time blocks. There are only a few studies which have considered this aspect.

Khan et al. [24] proposed a partial equilibrium linear model for optimal spatial and temporal synchronization of two streams of water and energy supply systems over their life-cycle. Dubreuil et al. [25] presented the evolution of different water and energy resource mix in supplying the demand at regional scale over the long-term multi-period planning horizon. They introduced the water-modelling module to the TIAM-FR model, a LP-based energy optimisation model. Parkinson et al. [26] presented a system analysis tool using multi-criteria analysis method for simultaneous long-term capacity expansion planning of water and energy technologies at national scale taking into account the Pareto optimal solution. Lv et al. [27] developed an optimisation model for planning water-energy nexus systems under uncertainty using an interval- fuzzy chance- constraint programming method. Saif and Almansoori [28] addressed simultaneous optimisation of water and energy supply chain problem. They applied a MILP model to optimise long-term capacity expansion and operation of an integrated system consisting of desalination, power and renewable energy power plants. They considered different desalination processes as well as the possibility of water and power transfer among different zones in the studied region.

These optimisation frameworks, however, have been developed at national or regional scales and naturally do not capture the complexities of the integrated system configurations. Therefore, they cannot be directly downscaled for the city scale. Additionally, they have taken into account the static operational constraints, which cannot adequately guarantee the validity and robustness of investment decisions as well as stability of the supply systems equipped with renewable energy technologies. In these cases, operational flexibility is a key factor which needs to be considered in the long-term planning due to its compatibility with inherent intermittency of this source of energy. This factor has been considered in optimisation studies only focusing on the operational aspect of the desalination plants/supply systems integrated with renewable energy sources. For instance, Smaoui and Krichen [29] developed a simulation-based algorithm for optimal energy management of a desalination unit powered by renewable energies. Hickman et al. [30] applied a mixed-integer quadratic constrained program for optimal operation of a system consisting of water, power, and co-production facilities. Al-Nory and Brodsky [31] presented a LP optimisation model for hourly scheduling of a desalination-based water supply system integrated with grid electricity and renewable energy sources. To address the operational flexibility, different time frames such as daily [32] and monthly [33] have been also taken into account.

To the best of our knowledge, there is no optimisation model incorporating short-term (hourly) operational constraints in long-term multi-period planning of the integrated water and energy supply systems at a city scale.

In urban areas located in arid and semi-arid regions, residential rooftop grid-connected photovoltaics (PVs) can potentially contribute to supplying sustainable and clean energy. In fact, the advantages of

being space-saving and efficient have turned them into an attractive option which competes with large-scale solar farms [34]. However, the highest possible level of their connection to the existing electrical grid is usually restricted to the dynamic penetration limit considered for grid stability purposes [19]. The intermittent penetration is due to the surplus PVs output which occurs during the day, when the generation is more than residential electricity demand.

There are many studies in energy sector which have addressed this issue. Among different approaches such as integrating several renewable energy sources, control strategies, using back up sources and storage technologies, the latter has attracted considerable attention owing to its ability to ensure grid stability and save excess generated energy for later use. For small applications, batteries have been suggested more than other storage technologies mainly because of their fast response, controllability and geographical independence.

Literature is rich in studies focusing on techno-economic analysis and optimisation of the PV-battery system. Using a battery-PV-simulation model, Linssen et al. [35] conducted a techno-economic analysis and determined the cost-effective configuration in various scenarios of economic and regulatory trends. Tomar and Tiwari [36] employed HOMER software for techno-economic analysis of grid-connected PV system in New Delhi considering feed-in tariff (FiT) and electricity rates. Hoppmann et al. [37] provided a literature review on the economics of batteries coupled with small-scale PV systems and investigated the role of different electricity price scenarios on the profitably of the systems with diverse capacities. Yang et al. [38] comprehensively reviewed the studies addressing the sizing of battery energy system and associated criteria as well as applied methods. Khalilpour and Vassallo [39] presented a review on the existing models for the optimal design of the PV-battery systems. They also developed a multi-period MILP model for their optimal configuration, size and operational decisions. Recently, Huang et al. [40] proposed a mathematical decision-making tool for optimal storage capacity in grid-connected PV systems and determined the relationship between storage capacities and the utilization rates of solar energy for efficient use of this energy source. The optimal power flows among different components of grid-connected PV –battery system have been also addressed previously. Sani Hassan et al. [41] developed a MILP model integrated with DER-CAM (distributed energy resources customer adoption model) software tool to optimally determine the power flows among different components of grid-connected PV–battery system. Grover-Silva et al. [42] addressed sizing and placement of distribution grid-connected battery systems taking into account operational strategies using a power flow distribution grid planning tool.

These studies have emphasised the benefits of using battery systems for removing the restriction of PV connection to the existing electrical grid. However, in practice, their application still highly depends on householders preference; even if the possibility of their usage is increased through rising residential electricity tariff, eliminating FiT or implementing incentive schemes to balance their initial cost [43].

An alternative approach to electricity storage is to introduce a deferrable load to the electrical grid that can assist to exhaust the surplus PV output at the time of its generation. In the context of the integrated water and energy management, this means each component of water supply system from the point of production to the end use is designed and operated compatible with the available renewable energy. This approach, therefore, leads to a higher PV installation capacity through exploiting its currently unused output. However, it also contributes to meeting a part of the intensive energy demand of desalination, which currently is a barrier to consider this technology as a sustainable long-term solution for water supply.

Align with our previous works [44,45], this study presents an interactive two-level MILP model for the multi-period long-term planning of a desalination-based water supply system driven by grid electricity and surplus PV output at a city scale. Considering short-term operational constraints, the optimisation model captures the intermittency of renewable energy and creates a hard-link between operation of the water supply system and available renewable energies in real-time (hourly) dispatch. The optimal strategic decisions are achieved based on this operational details. Simultaneously, the impact of the added water-related load to the existing electrical grid on the evolution of PV uptake capacity is quantified taking into account the householders' free will. This study, therefore, sheds a light on long-term effects of considering water and energy linkage in supply systems operation and planning and assists policy makers and engineers in making better informed decisions in both sectors. The proposed optimisation model can be used to quantify the interactive impacts of these distinct disciplines in urban demand supply when both sides of demand and supply are involved in the process of decision-making.

1.1. Contribution

This study addresses the multi-period long-term planning of a desalination-based water supply system driven by grid electricity and surplus PV output in urban areas. The main contributions of our work include: (1) short-term based strategic decisions for urban water and energy supply (2) detailed representation of a full day operation of the system (3) Ability to capture the inherent variability of renewable energy sources on hourly basis and incorporating it in both planning and operational decisions (4) Including the considerations of the spatial configuration of the integrated system (centralised vs. decentralised) in both operational and strategic decisions (5) Quantifying the effect of water and energy linkage on the evolution of the potential installation capacity of a renewable energy technology (PV system) temporally and spatially (6) Exploring the relation between economic performance of the integrated system and social and cost factors.

The remainder of the paper is as follows: Section 2 outlines the details of the proposed model. Section 3 describes the case study and alternative scenarios defined to indicate the capabilities of the model. Sections 4 presents the optimal strategic and operational decisions of the integrated system and their effects on the potential capacity of PVs uptake as well as the results of the sensitivity analysis. Lastly, the conclusion is presented in Section 5.

2. Methodology

In this study, the multi-period long-term planning of urban water and energy supply was addressed by developing an interactive two-level MILP optimisation model integrated with three tools of system advisor model (SAM), geographic information system (GIS) and Excel (Fig. 1).

In the first step, the temporal-spatial water and energy demands were forecast over the planning horizon within three time frames of yearly, seasonal (summer and winter) and hourly and for each distinct zone in the case study area. The latter was specified given the population (demand) distribution and service area of zone substations using ArcGIS 10 integrated with Excel analysis.

In the second step, the available resources for meeting the demands were determined. For this purpose, it was assumed that water demand was met by a desalination-based water supply system consisting of desalination plants, storage tanks and a pipeline network with different capacities. In zones equipped with a desalination plant, water demand could be provided from the combination of desalinated water (directly assigned from the plant), stored water and water transfer. In zones not equipped with a plant, demand could be satisfied only via water transfer from allowable (i.e. adjacent) zones. From energy point of view, it was presumed that residential and water-related electricity demands were satisfied by both grid electricity and residential rooftop grid-connected PV systems.

Grid electricity was delivered to the study area through distribution substations, which their maximum capacity was estimated in terms of their transformers' ratings [46]. The potential capacity of renewable

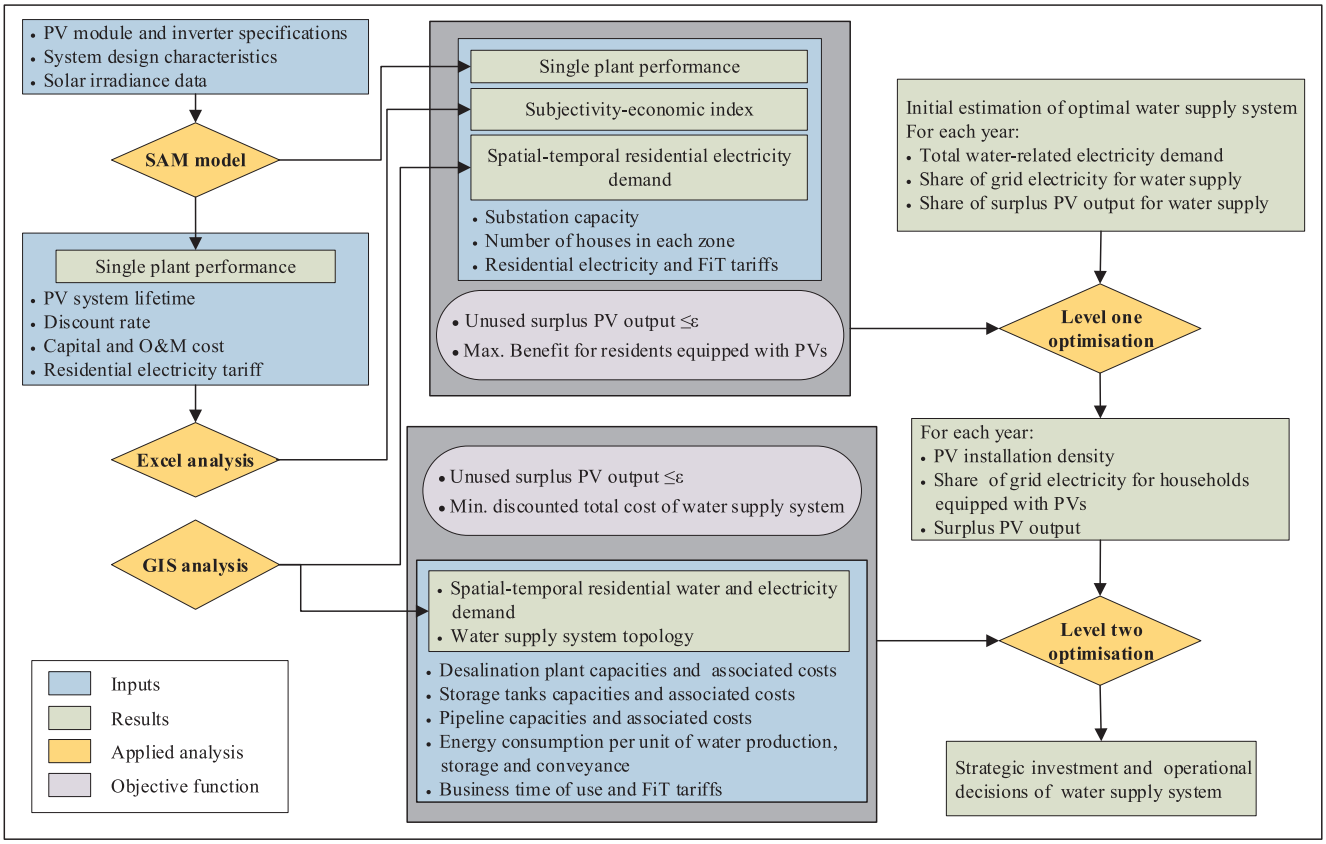


Fig. 1. Proposed optimisation model for an urban desalination-based water supply system driven by grid electricity and surplus PVs output.

energy for each year over the planning horizon was determined through optimisation model based on the generation of a selected PV system size and the number of houses in each zone as well as economic, subjectivity and technical criteria. The performance analysis of a selected PV system size was conducted using SAM model developed by the national renewable energy laboratory [47]. This model is equipped with a solar resource library from which the weather file for different countries and cities can be chosen. Other main SAM model input groups including system components technical specifications as well as system design and configuration are discussed in [48].

In the third step, the capital and operational and maintenance (O&M) costs of water and energy supply systems as well as electricity tariffs were estimated over the planning horizon. Lastly, the results of the previous steps were introduced to the developed two-level MILP model to achieve the optimal results.

In the following sections, the main aspects of the developed model are presented in more details.

2.1. Two-level mixed integer linear programming model

The optimisation problem was formulated as a MILP due to its flexibility and extensive modelling capability as well as the powerful solvers available commercially. The general mathematical representation of the problem is as follows (adopted from [18]):

$$\text{Minimise(or maximise)} \quad f(x, y) \tag{1}$$

$$\text{Subject to} \quad g(x, y) \leqslant b \tag{2}$$

$$\text{where} \quad X \in R \geqslant 0 \tag{3}$$

$$Y \in Z \geqslant 0 \tag{4}$$

In above equations, $f(x, y)$ expresses the objective function. X and Y are the vectors of continuous and integer (binary) decision variables, respectively. $g(x, y)$ is the vector of inequalities (and equalities) bounded by b, the vector of real constants. R and z are, in order, the sets of real numbers and integers.

The model consisted of a two-level of optimisation. In the first level, the objective function was to maximise the discounted total benefit that householders could achieve from their PV system in each year over the planning horizon. The economic benefit could be obtained via savings from avoiding grid electricity usage as well as revenues from FiT tariff by selling surplus system output to the grid. The outcomes of the level-one optimisation including the installation density, excess PV output and share of grid electricity in meeting residential electricity demand were then used as parameters in the next level of optimisation. At this stage, the objective function concerned the minimisation of the discounted total cost of construction, expansion and operation of the desalination-based water supply system.

In each level, objective function was subjected to a set of constraints tabulated in Table 1. Section I in the Supplementary document, presents the detailed mathematical formulation.

The model was coded into GAMS 24.3.1 [49], solved by CPLEX 12.6 and reached the final solution in two runs. In each run, both levels of the optimisation were solved in sequence. In the first run, the initial estimation of the optimal water supply system was achieved by allowing the whole area to be equipped with PV systems and relaxing the constraint regarding the unused surplus PV output. Thus, the results were determined without any restriction with regard to available renewable energy. The optimal solution in this run was obtained within a rather large relative optimality gap (< 10%).

The initial estimation for total water-related electricity demand, share of renewable energy and grid electricity in meeting this demand was then applied in the final run to adjust the PV installation density and give the optimal solution under the condition that no unused surplus output is allowed. This constraint led to the most compatible system operation with available renewable energy and contributed to

Table 1
Main constraints in each level of the MILP model.

Optimisation stage	Constraints
Level-one optimisation	Electricity balance for households equipped with PVs[a] Electricity balance for households not equipped with PVs[b] Energy resources capacity Maximum potential PV output Surplus PV output
Level-two optimisation	Water balance[c] Desalination plants capacities Storage tanks capacities Water pushed from desalination plant towards Storage tank Desalinated water storage Water flows Water-related electricity balance Electricity balance for households not equipped with PVs Energy resources capacity

[a] Electricity demand is supplied by PV system output and grid electricity. (Eq. A.2 in the supplementary document.)
[b] Electricity demand is met by surplus PV output and grid electricity. (Eq. A.43 in the supplementary document.)
[c] Water demand is satisfied by combination of water directly assigned from the desalination plant, stored in the tank and transferred from other zones. (Eq. A.18 in the supplementary document.)

stability of the electrical grid. The relative optimality gap criterion in this run was less than 0.001%.

The above-mentioned strategy for solving the problem was considered due to the interaction between the two levels of optimisation; however, it also assisted in confining the searching region and thus shortened the elapsed time of the program to find the final optimal solution. Below are the main decision variables of the model:

- Construction and expansion capacity and location of different components of the desalination-based water supply system including desalination plants, storage tanks and a pipeline network
- Hourly operational scheduling of water production, storage and transfer in each year
- Share of grid electricity and PV output in supplying water-related energy demand and total energy demand as well as its evolution
- Potential capacity of PVs uptake and its evolution

2.2. Effective criteria on the potential capacity of photovoltaic uptake

The evolution of the potential capacity of the PV uptake was determined through the optimisation process. In this regard, economic, subjectivity and technical criteria affected the value of this variable. The product of the economic and subjectivity indexes was introduced directly to the first level of the optimisation model, as a subjectivity-economic index, to specify the optimal PV installation density. The technical criterion restricted the capacity of installation to ensure the stability of the electrical grid. This criterion was considered as a constraint in both levels of the optimisation (except for the initial estimation of the optimal solution, Section 2.1). Each criterion is described in more details as follows:

Economic criterion: This criterion put the householder's economic priorities into perspective. Using the LC analysis, for each year during the planning horizon it determined whether it is economically worthwhile to uptake the PV system. The methodology used for this purpose was essentially based on [50]. First, the present total cost of a PV system that might be installed in each year during the planning horizon was ascertained by formulating the discounted cash flows of the capital and O&M costs (PC^{cap} and $PC^{O\&M}$) over the system's lifespan (Eqs. (5) and (6)):

$$PC^{cap} = \frac{CPV^{cap}}{(1+rr)^{n_t}} \tag{5}$$

$$PC^{O\&M} = CPV^{O\&M}.\frac{\left[\frac{(1+rr)^{PVL}-1}{rr}\right]}{(1+rr)^{n_t+PVL}} \tag{6}$$

where CPV^{cap} and $CPV^{O\&M}$ ($) are the capital and O&M costs of a specific size of a PV system, respectively; rr is a discount rate and n_t is the year of the installation (with $n_t = 0$, for the first year within the planning horizon). Finally, PVL (y) is the system lifetime.

The present values of the total cost of a PV system, were then annualised over the planning horizon using the equivalent annual annuity (EAA) method which uniformly distributes the cost of the system [50]. Accordingly, the LC of the PV system for each year that it might be installed was calculated considering the system output and efficiency. The latter was determined based on the annual degradation rate of the system.

In the next stage, the LC was compared to the electricity price tariff in each year. For a given year, should the latter be more than the LC, then it is economically beneficial to install the PV system. The outcome of this analysis was stored in a binary parameter (economic index). An economic index of 1 indicated that the associated criterion was met.

Subjectivity criterion: This criterion accounted for the householder's free will. Once the economic criterion for a given year was fulfilled, the subjectivity criterion specified what ratio of the householders would decide to install the PV system.[1] This criterion was defined as a positive parameter (subjectivity index) and could get any value between 0 and 1 assigned directly by the decision maker through interpreting statistics, forecasting trends and predicting householders behaviour. The latter could be affected by social awareness or their response to the peripheral components (like electricity markets [51]). The effect of the uncertainty of this parameter on the optimal solutions was specified by completing a sensitivity analysis.

Technical criterion: In this study, the issue of the intermittent surplus PV output was addressed by adding a deferrable load from the operation of the water supply system to the electrical grid. Given that the aim was to investigate the effect of this added load on the potential capacity of the PV uptake, the surplus output was allowed to be fed to the grid, which was not equipped with any storage technologies. Instead, the maximum potential PV uptake capacity was limited to the extent that no unused surplus output remains after meeting all energy demand.

2.3. Cost analysis

In this study, the discounted total cost of the water supply system was minimised to achieve the optimal strategic and operational decisions. Additionally, the total LC ($/m^3) was also considered as a metric to compare the optimal solutions in different scenarios. LC is an engineering economics index that gives the total unit cost of the product. According to Shahabi et al. [18], it is in fact "the real price at which a long-term contract would need to be negotiated in order for a project to breakeven in net value terms". Given the project lifetime and discount rate, LC was calculated in terms of the levelised capital cost (LCC ($/m^3)) and levelised operational cost (LOC ($/m^3)) as given by Eqs. (7)–(9):

$$LC = LCC + LOC \tag{7}$$

$$LCC = \frac{\sum_{t=0}^{WLT}\frac{CC_t}{(1+r)^t}}{\sum_{t=0}^{WLT}\frac{Q_t}{(1+r)^t}} \tag{8}$$

[1] In other words, the ratio of rooftops equipped with PVs to their total number.

$$LOC = \frac{\sum_{t=0}^{WLT} \frac{OMC_t}{(1+r)^t}}{\sum_{t=0}^{WLT} \frac{Q_t}{(1+r)^t}} \quad (9)$$

where CC_t and OMC_t ($) are, in order, the capital and operational costs of the water supply system occurring in the year t and Q_t is the total water production. r is the weighted average cost of capital (WACC) for water supplier and WLT is the lifetime of the project.

3. Case study and scenarios description

The north-western corridor of Perth, capital of Western Australia, was chosen as a case study to demonstrate the capabilities of the proposed model. This city has a population of more than two million people and due to its reliance on groundwater pumping and desalination, has had the highest energy intensity for water supply among major cities in Australia [52]. Currently, about half of water demand in Perth and surroundings is supplied by two seawater desalination plants (Perth and Southern) [53]. However, owing to urbanisation and the impact of climate change on groundwater resources, it has been suggested that a new 100 GL/y desalination plant, northern seawater desalination plant, needs to be constructed to meet the future demand in northern suburbs [54].

In this study, it was assumed that desalinated water is the only water resource supplying the demand in the case study area. Therefore, the problem was solved for three alternatives of desalination-based water supply system without considering the existing water supply infrastructures. Accordingly, business as usual (BAU) scenario considered a centralised desalination-based water supply system fuelled by grid electricity (as currently common practices). Two other scenarios were centralised and decentralised systems driven by both grid electricity and surplus PV output, named CGPV and DGPV, respectively. Regarding the level of water production in desalination plants, no hourly variations were considered for the fixed mode of operation in BAU and CGPV scenarios. In DGPV, however, the flexible operational scheduling was taken into account. Further details of different modes of operation can be found in [44].

The data collected for the case study composed of sets with both discrete values (i.e. water supply components capacities and cost data) and continuous values (i.e. distance and pumping elevation among allowable zones). Considering a planning horizon of 15 years, 2 representative days for seasons of summer and winter, 24 h of a day as well as 4 zones, the temporal datasets of water and energy demand included 2880 data points. For each zone in the study area, the parameters datasets like maximum capacities of substations, which were defined on yearly basis, included 60 data points. The yearly datasets such as cost data consisted of 15 data points, however, with regard to the grid electricity tariffs they increased to 720 data points considering 24 h of the representative day in each season. The following sections describe the case study data in more details.

3.1. Water and energy demands and supply systems

Water and residential energy demands in each of the four zones (Fig. 2) within the case study were estimated based on the demand per capita derived from [44] and the annual population growth of around 2.6% adopted from [55]. In this study, daily water demand was assumed to be the same in both seasons while residential energy demand varied. The average annual estimated water and residential energy demands in each zone are tabulated in Tables A.1 and A.2 of the supplementary document, respectively.

A seawater reverse osmosis desalination-based water supply system was considered to satisfy the required water. This technology was chosen since it is more likely to stay as the dominant desalination technology over the longer period due to its lower costs and energy consumption as well as ongoing technical advances [56]. The water

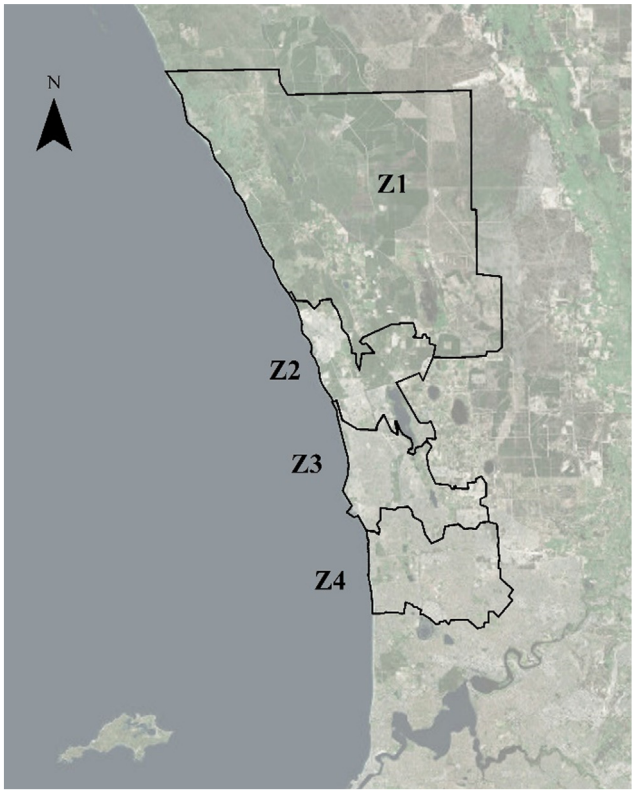


Fig. 2. Case study area and zone (z) boundaries.

supply system included desalination plants, storage tanks, and a pipeline network which their capacity/size(s) could be selected from different values summarised in Table A.3 of the supplementary document. Desalination plants and associated storage tanks were assumed to be located next to the ocean and in the relative centre of population, respectively. The number of days the plant operates divided by the total number of days in the planning horizon is the plant factor, which allows an ample time for preventive maintenance and unforeseen shutdowns. According to [57] it was set 0.85 for all desalination plants. The allowable capacities of the storage tanks were determined so that they could place minimum 2 h and maximum 3 days of water produced in the associated desalination plants, respectively. Similar to [44] no initial stored water was considered at the beginning of the planning horizon.

For BAU and CGPV scenarios, water could be transferred within zone 1 equipped with a large desalination plant or from this zone to others. This zone was selected for the location of the large capacity plant due to the land constraint in urban areas. In DGPV, water transfer could occur within any zone equipped with a desalination plant or among allowable (adjacent) zones. The details of the distance and pumping elevation differences were based on our previous study [44] (supplementary document, Tables A.4 and A.5). For water production, the average specific energy demand of 4 kWh/m^3 was considered according to [58]. The specific energy demand for water transfer was estimated based on [59].

The energy demand in the case study area was supplied by both grid and PVs (as described in Section 2). The maximum grid electricity that could be delivered to each zone was dictated by substation capacities. The associated specifications of the transformers in each substation were adopted from [60]. Table A.6 in the supplementary document presents the maximum capacity of the substations estimated in each zone over the planning horizon.

The performance of a single 4 kW PV system for 8760 h of a year was simulated using SAM 2016.3.14, since it was the most common system size used in the case study area [61]. SAM model input date was

Table 2
The model statistics for each scenario in the final run.

Scenario	No. of constraints	No. of total variables	No. of continuous variables	No. of binary variables	No. of iterations	Elapsed time (s)	Relative optimality gap (%)
BAU	126,551	56,381	56,051	330	5,746	10	–[a]
CGPV	126,551	59,261	58,931	330	5,076	11	–
DGPV	126,491	106,571	105,791	780	10,161,396	10,090	0.001

[a] The problem was solved to the optimality.

based on our previous study [44] (supplementary document, Table A.7). In this study, the effect of the climate change on PV system performance over the planning horizon was not taken into account.

Considering that the studied area is mainly occupied by single-level houses, the number of existing rooftops in each year was determined according to the estimated population and the number of people in each house based on [62]. In addition, for any given year where the economic criterion for PV system uptake was met, the subjectivity-economic index was assumed to be 1 corresponding to the highest potential capacity of PV uptake. In other words, the whole householders would decide to uptake PV system (subjectivity index = 1) if it was economically beneficial.

3.2. Cost data

All water and energy related costs were converted to 2017 Australian dollars ($) in real terms based on the appropriate exchange rates adopted from [63].

The capital cost of a 4 kW PV system was set 4230 ($) [64] and assumed to drop at a rate of 1.5% per annum over the planning horizon [65]. The system O&M cost is usually set as a percentage of the capital cost and was assumed to be 1% based on [50]. Considering that the useful lifetime of the inverter is less than PV modules (10–15 years of the inverter lifetime against 20 years (or more) of PV modules lifetime [50]), it needs to be replaced after 10 years since its installation. The associated cost was estimated at 5% of the total installed cost of the system based on [66]. The rate of discounting the cost and savings/revenues of the PV system usage in residential sector was considered 6.64% [67].

The price of grid electricity (for business and residential sectors) and FiT tariff ($0.07135/kWh) were adopted from [68,69]. For grid electricity, the price followed the time of use tariff structure for business sector (water supplier) consisting of fixed and variable electricity supply charge. The fixed electricity charge was set for each day of energy consumption while variable electricity charge was defined in terms of the amount of electricity consumption. No increase in electricity price was assumed over the planning horizon according to [65], which projected the relatively constant electricity retail prices per unit of electricity usage for the period between 2020 and 2037 for the south-west interconnected system (the electricity network in Perth and other regions in the south-west corner of Western Australia). It was explained that the cost of large-scale generation certificates under the large-scale renewable energy target will assist to stabilise the electricity prices in this network [65]. Similarly, the FiT tariff price was assumed to be the same in real terms over the planning horizon as most of the FiT tariffs have been discontinued by the Council of Australian Governments since 2012 and replaced with FiT tariffs with much lower rates. In fact, the savings achieved by avoided grid electricity cost have been the more effective motivator to increase the level of installations [65]. It was assumed that the structure of tariffs for both sectors remains constant over the planning horizon.

The capital and O&M costs of desalination plants were estimated based on [57]. For storage tanks and the pipeline network, the associated costs were taken from [70,71] and [18], respectively. Considering that the cost of water supply electricity usage varied by the source of energy (grid vs. renewable electricity), this cost component was excluded from O&M cost and calculated directly by the optimisation model. The breakdown of the total costs for water supply components was similar to our previous study [44]. Table A.3 in supplementary document presents the capital and O&M costs of different design capacities of desalination plants and storage tank as well as the unit-installed cost of the pipeline with various diameters. The pumping electricity cost associated with O&M cost of transferring water was calculated separately by the model.

Lastly, in order to calculate the discounted total cost of the water supply system, the real WACC of 5.1% was considered for water supplier adopted from [72].

4. Results and discussion

The two-level optimisation model was solved to the relative optimality criterion of 0.001% using the approach described in Section 2.1. The mathematical formulation was originally developed for DGPV scenario, and for other scenarios, the required constraints were added or modified to address their specific requirements. The statistics of the model for each scenario are summarised in Table 2.

The number of variables in BAU and CGPV scenarios was relatively the same and it increased significantly in DGPV scenario. The reason can be explained from two aspects. First, the location of siting the desalination plant in BAU and CGPV scenarios was limited to zone 1 while to find the best possible decisions in DGPV scenario all probable locations for siting water supply components needed to be taken into account. Second, the difficulty of finding the solution in presence of a wide range of time scales (in this case, yearly, seasonal and hourly), did not exist in BAU and CGPV scenarios but in DGPV. This is due to the constraint considered for the fixed operation in centralised scenarios, which limited the feasible searching region and led to the same optimal value for the majority of the operational-related variables on yearly basis. In contrast, in DGPV, the optimal values in each of the introduced time scale needed to be explored due to the flexible operation of the system. Accordingly, the elapsed time of the programme in this scenario was considerably longer (around 1000 times).

Although the number of variables was much greater in DGPV scenario compared to BAU and CGPV, owing to applying the same structure of the model (i.e. the constraints indices set), the number of generated constraints remained relatively the same in all scenarios.

Lastly, as indicated in the last column of this table, the model was solved to the optimality in BAU and CGPV scenarios and met the selected relative optimality criterion in DGPV implying that CPLEX produces strong bounds for optimal integer solution.

4.1. Optimal strategic decisions

The optimal solution resulted in the discounted total cost of about $2495, $2413 and $2244 million, in BAU, CGPV and DGPV scenarios, respectively.

In both BAU and CGPV, zone 1 was equipped with a centralised desalination plant, which met the whole demand during 15 years planning horizon (Fig. 3a). However, CGPV showed $82,139,609 less accumulated total O&M cost of the system compared to BAU due to incorporating the renewable energy in its water-related energy mix (Table 3). On the other hand, the optimal solution in DGPV led to a

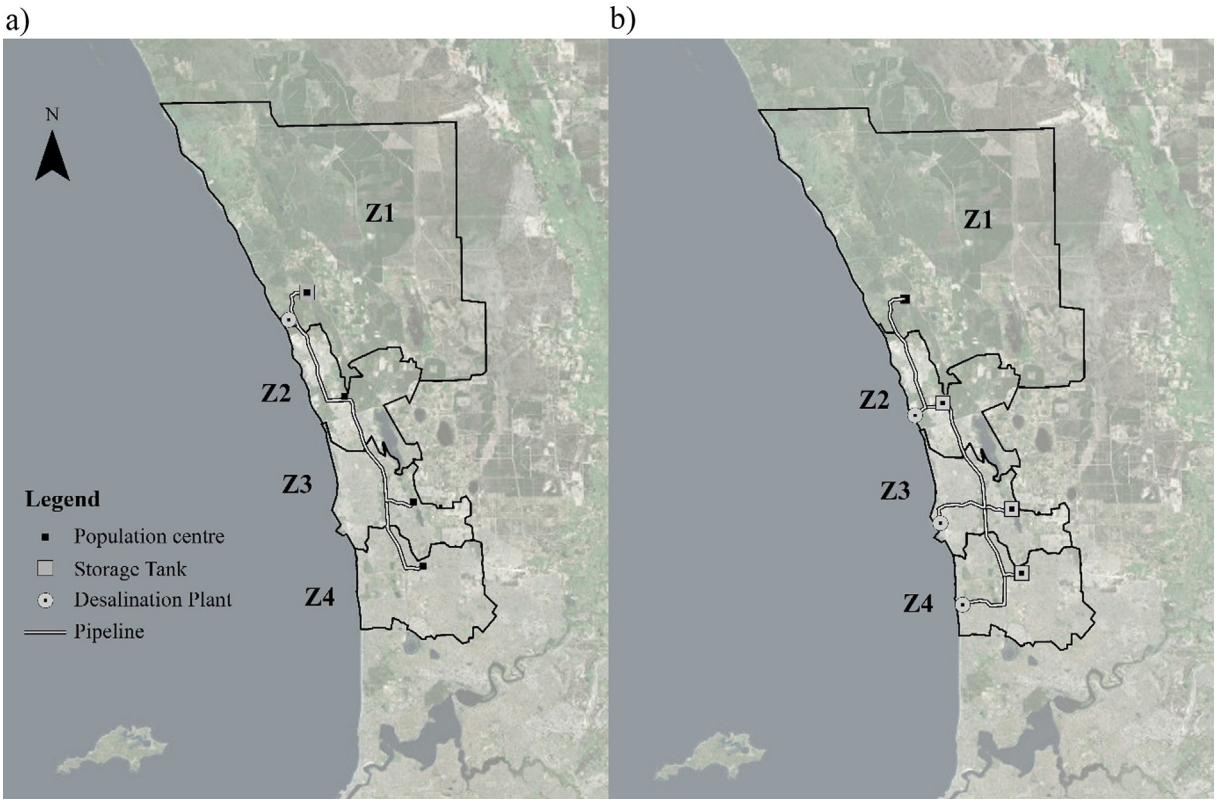


Fig. 3. Schematic view of the optimal locations for desalination plants, storage tanks and pipeline network by the end of planning horizon in (a) BAU and CGPV and (b) DGPV scenarios.

decentralised system configuration consisting of two desalination plants sitting in zones 2 and 4 in the first year and another plant in zone 3 in 2019 (Fig. 3b). In this scenario, the discounted capital cost was $8,035,261 more than centralised scenarios reflecting the absence of economies of scale of smaller desalination plants. However, its better economic performance can be explained by the discounted total system O&M cost achieved around $259,550,392 and $177,410,784 less compared to BAU and CGPV, respectively. The reason is associated with the operational flexibility of the system in this scenario, which resulted in taking the full advantage of the available renewable energy.

Overall, the results showed that the operational flexibility overweighs the spatial system configuration when there is the availability of

Table 3
Results of the optimal water supply system in different scenarios.

	Optimal scenarios		
	BAU	CGPV	DGPV
LC of the water supply[a] ($/m^3) and relative difference with BAU scenario	3.30/–	3.19/3.33%	2.96/10.3%
Absolute difference in discounted total cost of the water supply system with BAU scenario ($)	–	$82,139,609	$251,515,132
Breakdown of the discounted total cost ($)			
Desalination plants and storage tanks (capital)	$1,324,747,281	$1,324,747,281	$1,425,853,173
Pipeline network (capital)	$98,945,428	$98,945,428	$5,874,797
Desalination plants and storage tanks (O&M)	$1,007,580,082	$925,810,190	$808,417,775
Pipeline network (O&M)	$64,102,434	$63,732,717	$3,714,348
Capacity construction and/or expansion details			
Desalination plant location, capacity (m^3/day), construction year, expansion capacity (m^3/day), expansion year	Z1(280,000), year 2017, no expansion	Z1(280,000), year 2017, no expansion	Z2(80,000), year 2017, no expansion Z3(40,000), year 2019, no expansion Z4(120,000), year 2017, (40,000), year 2026
Storage location, capacity (m^3), construction year, expansion capacity (m^3), expansion year	Z1(20,000), year 2017, no expansion	Z1(20,000), year 2017, no expansion	Z2(10,000), year 2017, no expansion Z3(10,000), year 2019, no expansion Z4(10,000), year 2017, (10,000), year 2026
Pipeline (links)	Z1-Z2 Z1-Z3 Z1-Z4	Z1-Z2 Z1-Z3 Z1-Z4	Z2-Z1 Z2-Z3 Z4-Z3

[a] This economic metric has been calculated considering all components of the desalination-based water supply system including production, storage and distribution.

renewable energy. This could challenge the default belief that desalination systems always need to benefit the economies of scale [73] and provide a wider range of integrated system alternatives for the decision maker to address the long-term urban water demand.

The following section presents the optimal operational details of the water supply system in year 2028 for a representative zone to provide an insight to how short-term operational scheduling can influence the economic performance of the system in long-term horizon.

4.2. Optimal operational scheduling

The optimal operation of the water supply system followed the same logic in all zones and seasons in each scenario. However, there was a slight difference between the paradigm of the operation in summer and winter time. This difference is associated with the variation of available surplus PV output as a result of seasonal fluctuations of solar radiation as well as residential electricity usage profile. In [44], the effect of seasonal changes on the optimal operation of the desalination-based water supply system was discussed in details. In this section, the daily operational details of the system to supply the water demand in zone 4 and year 2028 during summer time is presented. The optimal solution for winter time can be found in Section III of the supplementary document.

In DCPV scenario, water demand in zone 4 was supplied within the same zone. At the point of production, the highest level of water was produced during the availability of surplus PV output, even though it was concurrent with the peak electricity rates period (Fig. 4a). In these hours, the extra water, around 50% of the production, was pushed for storage. At the point of demand, the stored water provided total water demand during the peak electricity rate hours when no renewable

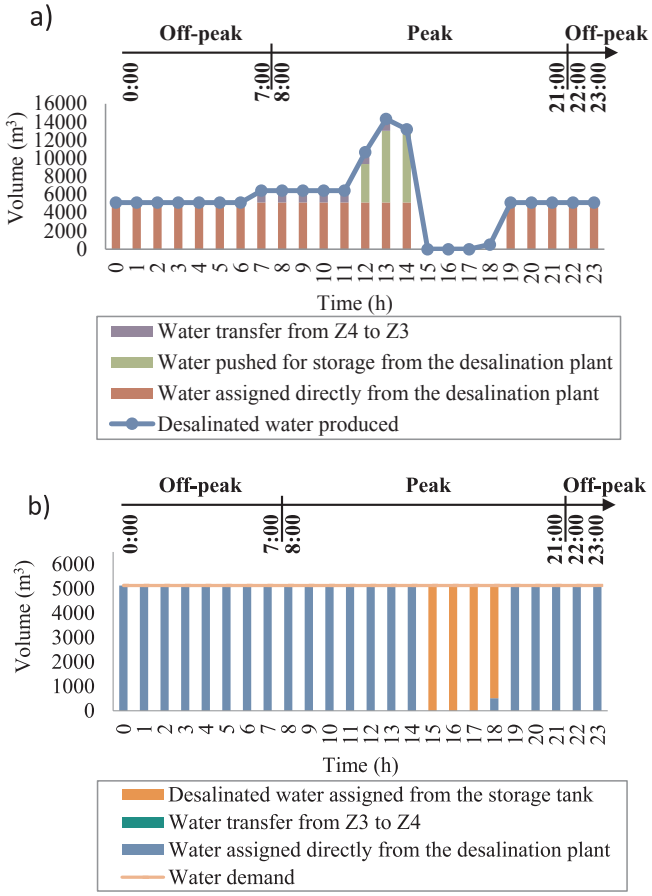


Fig. 4. The operational scheduling of the water supply system in DGPV scenario: (a) at the point production (b) at the point of demand.

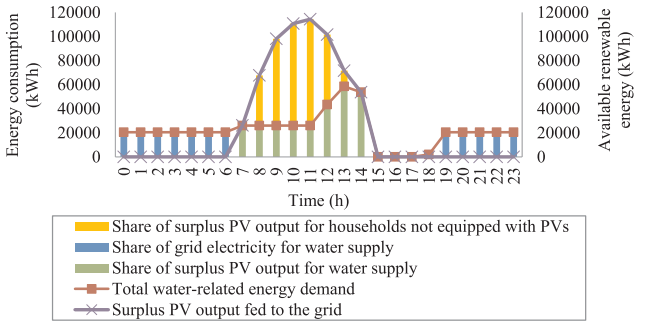


Fig. 5. Share of each energy source in supplying water-related electricity demand at the point of production in DGPV scenario.

energy was available (Fig. 4b). Considering that in this scenario zone 4 was equipped with a desalination plant, the water demand in other times of the day was supplied directly from the desalination plant. This operational scheduling led to the share of 53.53% of available renewable energy in water related energy mix in this zone (Fig. 5) corresponding to $20,461,995 electricity cost savings for water supply in summer time.

In CGPV, the point of water production for the whole study area (including zone 4) was located in zone 1 (Fig. 6). Although this resulted in a high water-related energy demand, due to the centralised configuration, the system had access only to surplus PV output in this zone. Thus, while the total available renewable energy was consumed, it only contributed to 10.34% of the total water-related energy demand resulting in $6,800,708 energy savings during summer (Fig. 7a). However, this scenario was still more profitable in long-term run over BAU, which did not include renewable energy in its energy mix (Fig. 7b). It is notable that the different amount of surplus PV output in zone 1 in CGPV and BAU scenarios (Fig. 7a and b) is related to the maximum PV uptake capacity which has been limited in each scenario to the extent that can be fully consumed by water supply system and households not equipped with PV systems (Section 2.2).

At the point of demand (zone 4), in both scenarios the demand was supplied by water transfer from zone 1 (Fig. 8). Therefore, the surplus PV output was not assigned to water supply and exhausted only by households not equipped with PVs (Fig. 9).

The significant effect of daily operational scheduling and spatial configuration of the system on its short-term economic performance reflects their important role in long-term planning the integrated water and energy supply especially when renewable energy sources are considered as a part of the energy mix (similar to GEP problems in energy sector [74]). According to the results, it is evident that disregarding these aspects, as it has been the case in a few multi-period long-term planning models developed previously for integrated systems (For

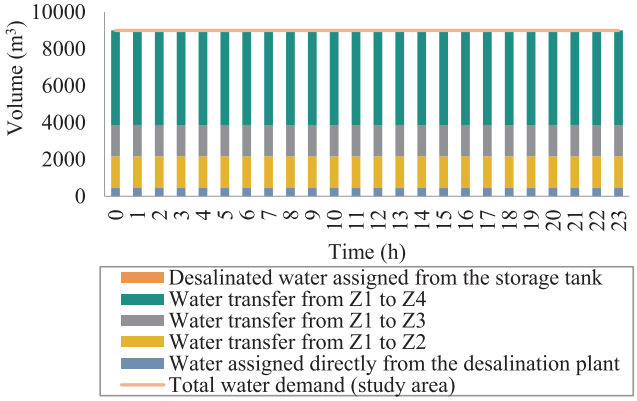


Fig. 6. The operational scheduling of the water supply system in CGPV and BAU scenarios at the point of production.

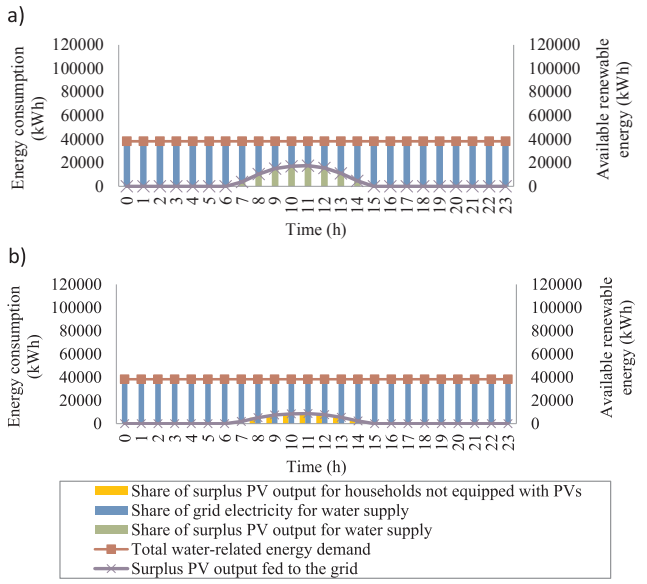


Fig. 7. Share of each energy source in supplying water-related electricity demand at the point of production in: (a) CGPV and (b) BAU scenarios.

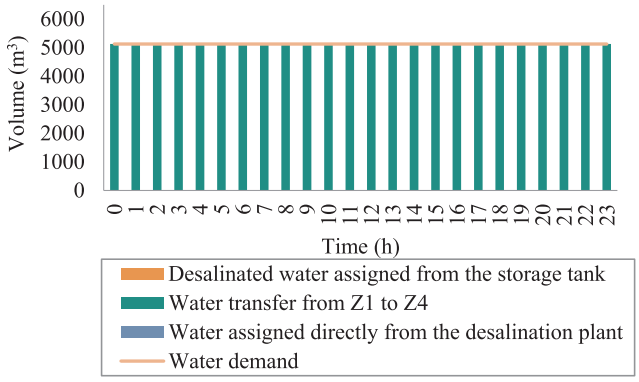


Fig. 8. The operational scheduling of the water supply system in CGPV and BAU scenarios at the point of demand.

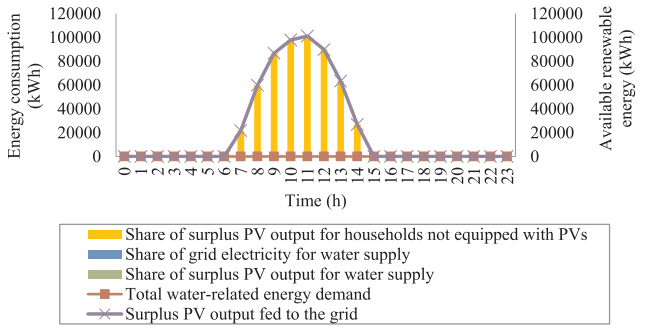


Fig. 9. Share of each energy source in supplying water-related electricity demand in BAU and CGPV scenarios at the point of demand.

instance Ref. [28]), may question the validity and robustness of their investment decisions to be implemented.

4.3. Potential photovoltaic uptake capacity and share of energy sources in water supply

The evolution of renewable energy share in supplying water-related energy demand as well as the potential PV uptake capacity in all scenarios are shown in Figs. 10–12.

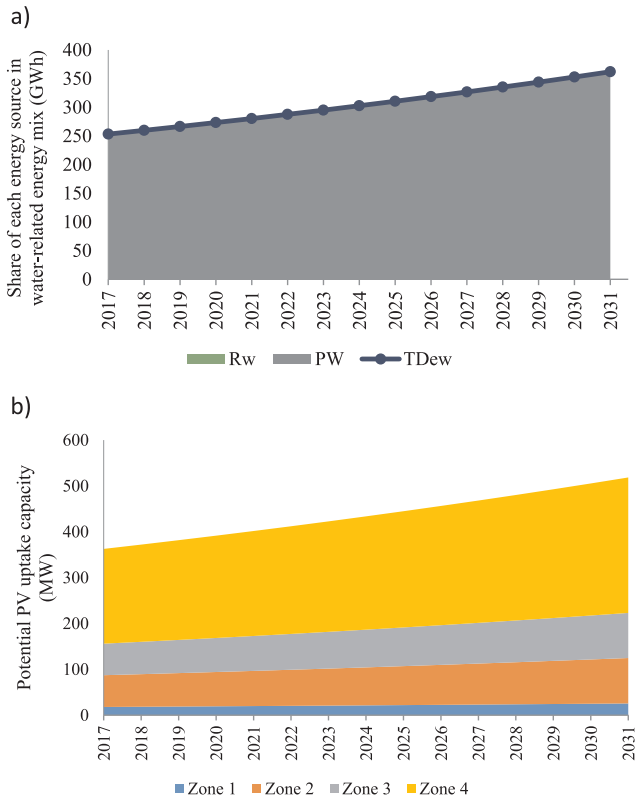


Fig. 10. BAU scenario: (a) share of each energy source namely renewable energy (Rw) and grid electricity (Pw) in supplying total water-related energy demand (TDew) over the planning horizon and (b) the evolution of potential capacity of PV uptake in the case study area.

As mentioned in Section 3, in BAU scenario, water supply system was operated in fixed mode regardless of the available renewable energy. Consequently, total water-related electricity demand was supplied by grid electricity (Fig. 10a) and the potential PV uptake capacity was limited to the extent that the surplus PV output can be fully used at the time of its generation by households not equipped with PV systems. In this scenario, the average total potential PV uptake capacity in zones 1, 2, 3 and 4 was in order, 21,549.3, 83,350.3, 82,760.3 and 248,657.1 kW corresponding to average installation density of 49.9% for all zones in each year over the planning horizon (Fig. 10b).

In CGPV scenario, the available renewable energy in zone 1, where the centralised desalination plant was located, contributed to meeting water-related energy demand. The added load, led to the PV installation density of 100% in this zone and 21,630.8 kW higher potential PV uptake capacity on average in each year over the planning horizon compared to BAU scenario, (Fig. 11b vs. Fig. 10b). Owing to the high water-related energy demand in this zone (associated with supplying the whole water need in the studied area), the total available surplus PV output was consumed by water supply system, although it could only meet 6.4% of its demand (Fig. 11a). In fact, in this scenario, the share of renewable energy in water supply was restricted by the number of existing rooftops estimated in this zone over the planning horizon. The potential PV uptake capacity and its installation density in other zones (namely, 2, 3 and 4) were similar to BAU scenario (Fig. 11b), bounded by the available load from the households not equipped with PVs.

In DGPV scenario, around 36% of the water-related energy demand in the case study area was met by renewable energy in the first two years which then increased to 43% in 2019 (Fig. 12a), as a result of the new desalination plant constructed in zone 3 (Section 4.1, Table 3). Over the years between 2019 and 2026, on average 42% of the demand was provided by this source of energy which then raised slightly to 46% (Fig. 12a) due to the expansion of the desalination plant located in zone

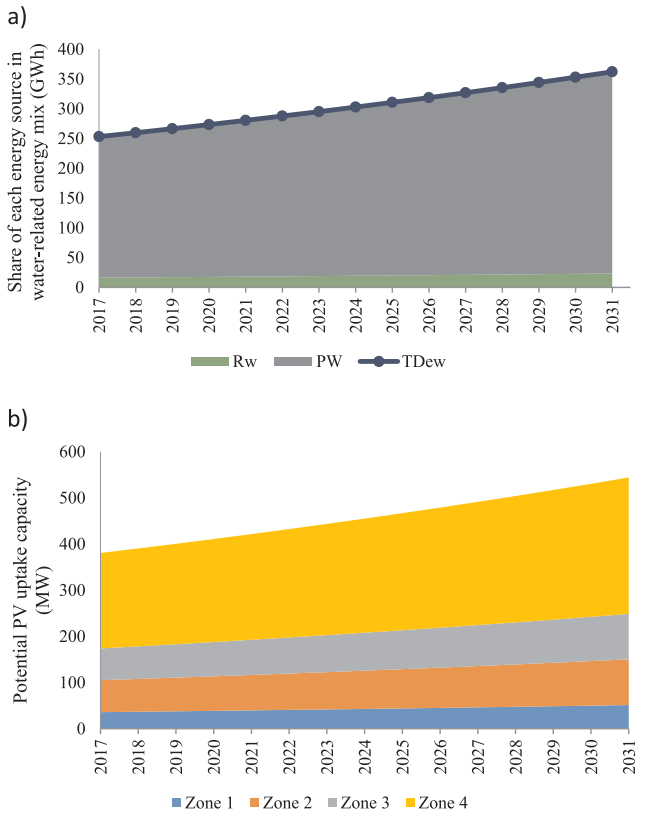


Fig. 11. CGPV scenario: (a) share of each energy source namely renewable energy (Rw) and grid electricity (Pw) in supplying total water-related energy demand (TDew) over the planning horizon and (b) the evolution of potential capacity of PV uptake in the case study area.

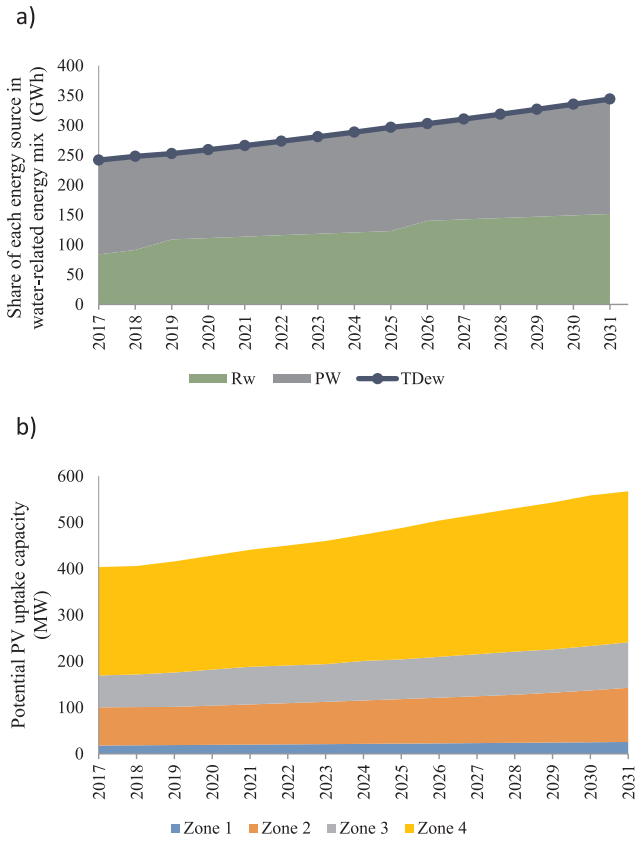


Fig. 12. DGPV scenario: (a) share of each energy source namely renewable energy (Rw) and grid electricity (Pw) in supplying total water-related energy demand (TDew) over the planning horizon and (b) the evolution of potential capacity of PV uptake in the case study area.

4 (Section 4.1, Table 3). In this scenario, the level of the contribution of the renewable energy to water supply was affected by the level of available surplus PV output as well as water demand in each zone, which to some extent dictated the required water-related energy demand. Compared to both BAU and CGPV, the added load from water supply system in zones 2, 3 and 4, in order, led to 11,990.6, 1,729.1 and 29,045.4 kW higher potential PV uptake capacity on average in each year over the planning horizon (Fig. 12b vs. Fig. 10b and 11b). Accordingly, as shown in Fig. 12b, this scenario resulted in the average total potential PV uptake of 21,549.3, 95,340.9, 84,489.4 and 277,702.4 kW corresponding to the average installation density of 49.9%, 57.1%, 51% and 55.7%, in zones 1, 2, 3 and 4 respectively, over the planning horizon.

In general, adding water-related energy load to the existing electrical grid results in a significant increase in uptake capacity of a small-scale renewable energy technology (rooftop PVs) which best fits the built environment. This approach is especially applicable in land-restricted urban areas as it assists to explore the potentials and exploit the already existing clean energy sources.

Additionally, the results clearly demonstrate the effect of considering water and energy linkage in the optimal solution in distinct periods over the planning horizon as opposed to the previous studies quantifying this interaction for a single/several years in the future, for instance across their supply systems lifecycles [24] or within technological options [25]. In practice, considering this aspect in long-term planning models can lead to making better informed decisions as it captures the fluctuations of different parameters (demand, cost, markets, etc.) over time and provides a holistic view of the consequences of decisions made in each sector and their impacts on the other.

4.4. Sensitivity analysis

In this section, the sensitivity of the optimal strategic and operational decisions towards different values for the subjectivity index and O&M cost ratios of flexible to fixed operational scheduling is shown in Fig. 13a and b, respectively.

In respect to potential PV uptake capacity, it was assumed that in case the economic criterion was met, all householders in the studied area would be interested in uptaking PV system (subjectivity index = 1, Section 3.1). Therefore, the only limiting factor was the available load on the electrical grid which could exhaust the surplus electricity at the time of its generation. This assumption, therefore, led to the highest potential PV uptake capacity. To evaluate the sensitivity of the optimal solutions towards different values for the subjectivity index, the model was run for four different values, namely 0.75, 0.5, 0.25 and 0. These values represented the ratio of the householders interested in uptaking PVs in case it is economically beneficial (Section 2.2). The subjectivity index of 1 was considered as the reference case.

As expected, the variation in subjectivity index values did not affect the optimal solutions in BAU scenario considering that in this scenario water supply system was operated in fixed mode independent from the level of available renewable energy (Fig. 13a). Similarly, the results indicated high resilience to changes in the subjectivity index in CGPV scenario due to the fact that only the renewable energy available in zone 1 contributed to supplying water-related energy demand.

In contrast, the optimal decisions in DGPV scenario showed high sensitivity towards the subjectivity index values. Compared with the reference case, the optimal solutions associated with the subjectivity values of 0.75, 0.50 and 0.25, led to $11,525,794, $45,750,176 and $116,816,743 higher discounted total cost of the water supply system,

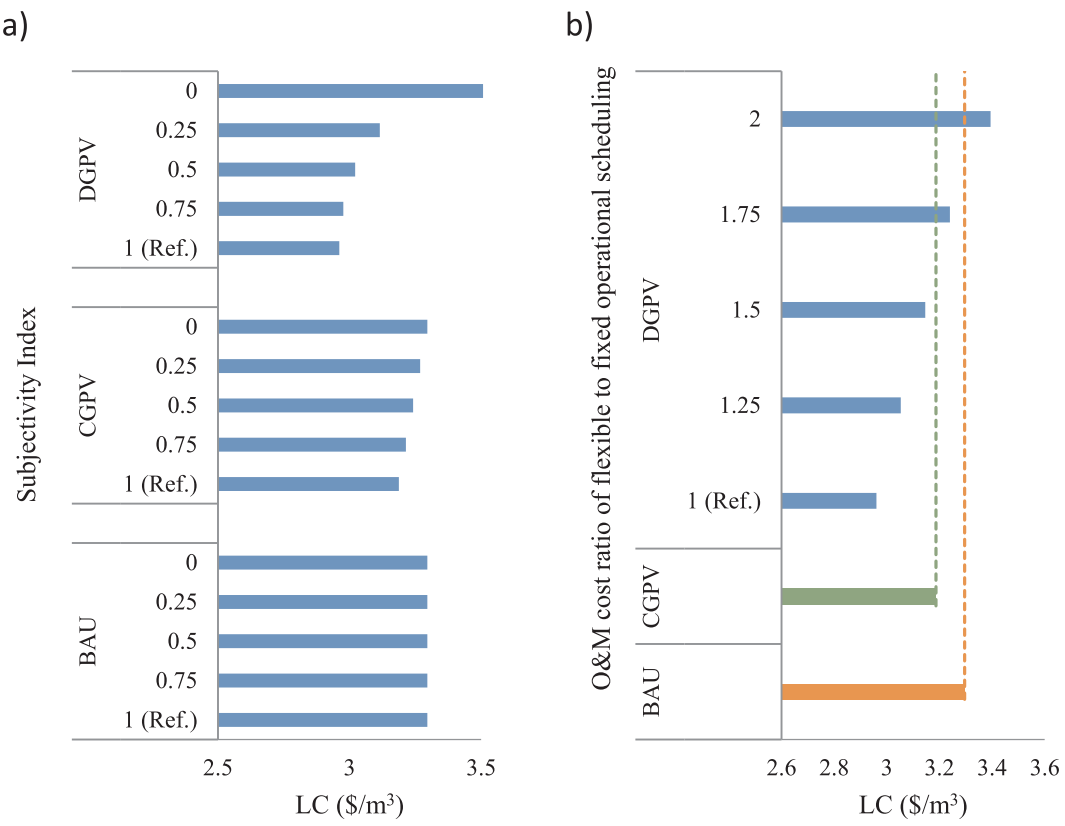


Fig. 13. Sensitivity of the LC of the optimal water supply system in BAU, CGPV and DGPV scenarios for different values of: (a) subjectivity index and (b) O&M cost ratio of flexible to fixed operational scheduling.

respectively. In the case of subjectivity index value of 0%, the LC of the system was 19.5% higher compared to the reference case. The discounted total cost of the system in this case was $185,283,888 and $267,423,497 greater than BAU and CGPV scenarios, respectively. This suggests that in the absence of renewable energy, it is not economically beneficial to choose decentralised over centralised water supply system and highlights the importance of developing policies favouring higher installation of PVs (such as incentive programs) in the case study area.

The sensitivity of the optimal decisions in regard to the O&M cost associated with different modes of operation was also investigated. In this study, the solutions in the reference case were achieved regardless of the effect of operational mode of water supply system on this cost component. The reason was due to the fact that the operational mode commonly used in many desalination plants worldwide is fixed and therefore the available O&M cost data are mainly based on this operational scheduling. To investigate the probable adverse effect of the flexible operational mode on water supply infrastructures in terms of maintenance, repair and replacement, the optimal decisions were obtained for four different ratios of O&M cost of flexible operational mode (applied in DGPV scenario) to O&M cost of fixed operational mode (implemented in BAU and CGPV scenarios). The ratio of 1 was considered as the reference case.

Fig. 13b indicates the results of the sensitivity analysis achieved in the ratios of 1.25, 1.5, 1.75 and 2. The results showed high sensitivity towards probable increase in the O&M cost in flexible operational mode. Up to the point where the O&M cost increased to 1.5 times of the fixed operational mode, DGPV scenario had still the economic preference over both BAU and CGPV scenarios. However, in the ratio of 1.75, the discounted total cost of the water supply system was $41,822,339 higher than CGPV scenario and in the ratio of 2, it was $76,802,228 greater than BAU scenario. This implies that in order to select the best scenario in real world cases, the possible consequences of flexible mode of operation in terms of extra O&M cost need to be investigated more accurately by long-term running of water supply system components in laboratory and pilot scales.

5. Conclusion

This paper presented an interactive multi-period long-term planning model for integrated urban water and energy supply system incorporating short-term operational constraints. Grid electricity and surplus output from grid-connected residential rooftop photovoltaics were considered as energy sources for a desalination-based water supply system. Accordingly, the optimal strategic decisions of the system were achieved considering fluctuations of the available renewable energy. The findings of the study include the optimal operational scheduling and capacity expansion planning of the water supply system as well as the evolution of photovoltaic uptake capacity.

Three key insights were obtained through applying the model to the case study of Perth, Australia. From water sector point of view, it was observed that the operational flexibility and decentralised configuration of the water supply system resulted in 10.3% less levelised cost, accounting for around $251,515,132 discounted total cost saving over the planning horizon, compared to the currently common centralised system. This can question the default belief that desalination systems always need to benefit the economies of scale and provide the decision maker with a wider range of integrated system options to address sustainable water supply when there is the availability of renewable energies.

From energy point of view, it was found that adding flexible water-related energy load to the exiting electrical grid at the time of surplus photovoltaic generation significantly increased the potential uptake capacity of this technology over the long-term planning horizon (on average 42,765.1 kW annually). Therefore, the model is able to capture the linkage of water and energy and is especially useful when it comes to the efficient consumption of already existing resources in urban

areas.

Finally, it was evident that the lower level of photovoltaic uptake capacity and higher probable imposed operational and maintenance cost of flexible mode of operation increased the levelised cost of the decentralised system up to 19.5% and 14.6%, respectively. This implies the importance of adopting policies favouring higher installation of renewable energy technologies as well as the accurate estimation of the operational cost component in selecting the best alternative for sustainable supply.

This research considered desalination technology as the only water supply option in the study area. The integrated system, however, can be expanded by incorporating the existing surface water resources as well as the effects of climate change. In this vein, considering the studies dealing with predicting the hydrological data [75] or measuring solar radiation [76] can be beneficial. The other research challenge that the future works need to address is to also include more detailed technical constraints associated with the real-time water and energy systems dispatch.

Acknowledgement

Ms. Negar Vakilifard would like to thank Murdoch University for supporting this research by awarding Murdoch International Postgraduate Scholarship (MIPS).

Appendix A. Supplementary material

Supplementary data to this article can be found online at https://doi.org/10.1016/j.apenergy.2018.10.128.

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